
Frozen Regime Stress Tests for Cryptocurrency Direction Prediction

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GitHub repository

1 Introduction

Main result. On 1,488 BTC/ETH forecast origins from July 2026 that were not used for development, the frozen CNN–LSTM reached macro-F1 0.409 and accuracy 0.489. Its historical test macro-F1 was 0.429 ± 0.004 across three seeds. When I trained on one volatility regime and tested on the other, macro-F1 fell by 0.081 ± 0.005 (low→high) and 0.057 ± 0.005 (high→low). The model learned useful directional patterns, but those patterns did not transfer evenly across volatility regimes. Macro-F1 is the unweighted mean of the three class F1 scores; accuracy is the overall correct fraction.

This project predicts whether the next-hour close return for BTCUSDT or ETHUSDT is *down*, *flat*, or *up*, then asks how the predictions change with volatility. The task is difficult because crypto returns are noisy, the classes are imbalanced, and market behaviour changes over time. A convolutional layer can summarize short local patterns, while an LSTM can carry information across the full input window. The study tests predictive associations, not causality or trading profitability.

1.1 Related Work

Fischer and Krauss benchmarked LSTMs against memory-free classifiers for equity direction and found that performance varied across sub-periods [1]. Nelson et al. used technical indicators with an LSTM for stock movement classification [4], while Ghosh et al. compared LSTMs and random forests on out-of-sample intraday directions [2]. For cryptocurrency data, Zhao modeled short-horizon movement from trade-by-trade sequences [5]. Guo et al. combined multiscale residual convolution and LSTM components for Bitcoin price forecasting [3]. Together, these studies motivate sequence models and strong simple baselines. My evaluation adds causal volatility slices, paired cross-regime transfer, three training seeds, and a future month fixed before it was examined.

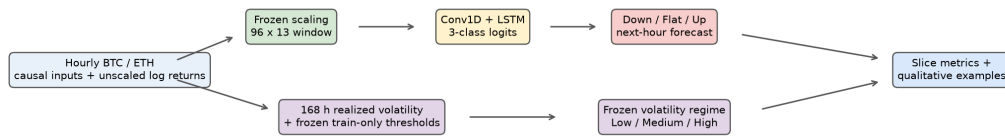


Figure 1: Two frozen paths share each forecast origin: a 96×13 sequence produces a three-class prediction, while trailing 168-hour realized volatility assigns a train-thresholded regime. Their outputs meet only during evaluation, preventing the next-hour outcome from defining its own regime.

2 Data Processing

Binance public hourly OHLCV klines cover July 1, 2023 00:00 UTC through June 30, 2026 23:00 UTC: 26,304 rows per asset (52,608 total). Timestamps were converted to UTC, sorted, deduplicated, and required to be hourly contiguous. Thirteen causal features were computed: simple and log returns, high–low range, close–open return, volume change, RSI(14), MACD(12,26,9), Bollinger position and width(20), 24-hour volatility, and 24-hour volume z-score. A sample at forecast origin t contains the preceding 96 hourly feature rows through t ; its label uses the close return from t to $t + 1$. The training-only median absolute return, 0.2357%, defines down ($r < -0.2357\%$), flat, and up ($r > 0.2357\%$).

Chronological 70/15/15 partitions yielded 36,584 training, 7,698 validation, and 7,702 historical test windows. Training class counts were 8,838/18,230/9,516 (down/flat/up); historical test counts were 2,061/3,703/1,938. Feature means/scales, label threshold, and model selection were fitted using training or validation data only; no window crosses a symbol or split boundary.

Genuine example. At BTCUSDT 2026-06-30 22:00 UTC, raw open/high/low/close/volume were 58,639.99/58,806.91/58,513.80/58,607.99/274.43078. The engineered one-hour return was -0.000546 , RSI(14) was 38.365, and 24-hour volatility was 0.003793. The next close was 58,624.71, a return of $+0.000285$; because its magnitude was below the frozen threshold, the resulting class was flat.

Volatility regimes. For each asset, trailing realized volatility at t is the population standard deviation of 168 hourly log returns available through t . The low/high cut points are the training-only 33rd/67th percentiles: BTC 0.00390/0.00512 and ETH 0.00520/0.00674. Values at or below the lower cut point are low; values above the upper cut point are high; the rest are medium. Per-asset thresholds avoid using asset identity as a volatility proxy.

3 Architecture

The 18,067-parameter PyTorch model applies same-padded Conv1D ($13 \rightarrow 32$, kernel 5), ReLU, batch normalization, and 0.15 dropout to all 96 steps. A one-layer LSTM with 48 hidden units feeds the last hidden state through 0.20 dropout and a $48 \rightarrow 3$ linear logit head. CrossEntropyLoss uses inverse-frequency training weights $w_c = N/(3n_c)$ and consumes logits directly; Softmax is applied only at inference. Adam (10^{-3}), batch size 256, gradient clipping at 1.0, at most 12 epochs, and validation-loss patience 3 were used. Seeds 42–44 quantify variation; minimum validation loss selected seed 43 before prospective access.

4 Baseline Models

All baselines use identical windows and labels. *Majority* predicts the most frequent training class (flat). *Momentum* maps the last observed return through the frozen class threshold. Class-balanced multinomial *logistic regression* uses the last 13-feature vector. These represent a no-signal prior, a parameter-free heuristic, and a fast memory-free learned comparator.

5 Quantitative Results

Table 1 reports the two final evaluation periods. Historical CNN values are mean $\pm$ population standard deviation over seeds 42–44; deterministic baselines and the frozen prospective seed are single evaluations. I use macro-F1 as the primary metric because flat is the largest class.

Table 1: Overall performance on the historical test ($n = 7,702$) and prospective July 2026 ($n = 1,488$). Every value comes from the saved genuine result JSON.

Model	Historical test		Prospective July	
	Macro-F1	Accuracy	Macro-F1	Accuracy
Majority	0.216	0.481	0.243	0.573
Momentum	0.380	0.421	0.356	0.447
Logistic regression	0.429	0.480	0.380	0.529
CNN-LSTM	0.429 $\pm$ 0.004	0.450 $\pm$ 0.005	0.409	0.489

On the historical test, the CNN matched logistic regression on macro-F1 but not accuracy. Its recalls were 0.383/0.514/0.398 (down/flat/up), more balanced than logistic regression’s 0.271/0.656/0.366. In July, CNN recalls were 0.368/0.627/0.244: up recall weakened, but the CNN still had the best macro-F1. The majority baseline obtained high accuracy mainly because 853 of 1,488 July labels were flat, which is why accuracy alone gives an incomplete picture.

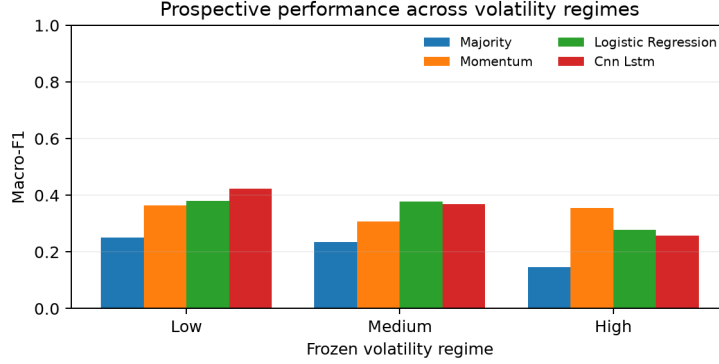


Figure 2: Prospective macro-F1 under frozen regimes. CNN performance fell from 0.424 in low ($n = 1,005$) to 0.368 in medium ($n = 408$) and 0.258 in high volatility ($n = 75$). The high-regime estimate is especially uncertain because that slice is small.

The CNN declined monotonically across the prospective regimes, although not every baseline did. This is evidence of sensitivity to volatility, not a universal causal effect. Historical global CNN macro-F1 was 0.409/0.413/0.379 for low/medium/high ($n = 2,033/3,240/2,429$). The paired transfer experiment showed the same weakness: CNNs trained only on low regimes scored 0.381 ± 0.006 on low versus 0.300 ± 0.001 on high, a paired change of -0.081 ± 0.005 . High-trained CNNs scored 0.409 ± 0.005 on high versus 0.352 ± 0.001 on low, a change of -0.057 ± 0.005 . Logistic changes were -0.063 and -0.071 . In this study, these paired drops provide the strongest evidence that performance depends on the training regime.

6 Qualitative Results

I selected examples using a fixed rule: the highest-confidence correct prediction and error in each non-empty July regime. In low volatility, the strongest correct case was BTC flat at 88.6% confidence (July 18 11:00 UTC), while the strongest error predicted flat at 87.2% for an actual BTC down move (July 25 04:00). Medium-regime examples repeated this pattern: an 85.0%-confidence correct flat and an 81.7%-confidence false flat. In high volatility, confidence was lower: an ETH down prediction was correct at 50.1%, while another down prediction missed an up move at 48.9%. This small set illustrates the same pattern as the aggregate results: the model can be confidently biased toward flat in quieter periods and struggles to separate directions during rare high volatility.

7 New Data

Before looking at July, I trained the models, selected seed 43 by validation loss, and froze the CNN weights, baselines, scaling, label/regime thresholds, feature order, and exact July interval. I committed that state before fetching July data, then ran the prospective command once. It used June 24–30 solely as causal context and scored exactly 744 origins per asset from July 1 00:00 through July 31 23:00 UTC; the August 1 00:00 close resolves the last label. The 1,488 targets had 307/853/328 classes and 1,005/408/75 low/medium/high regimes. After seeing the July scores, I did not change the model, retune it, or run the prospective evaluation again. The CNN’s macro-F1 declined from 0.429 historically to 0.409 in July, so the historical performance did not fully carry over. However, the CNN still had the best July macro-F1 among the tested models. This supports a narrow claim about next-hour classification, not profitability.

8 Discussion

The hybrid model did not dominate every comparison: it only tied logistic regression on historical macro-F1, and the majority baseline won July accuracy. Its advantage was more even recall across the three classes, which gave it the best July macro-F1. At the same time, both transfer directions and the prospective regime slices showed that this advantage was unstable. Low-trained models lost flat recall in high volatility, while high-trained models over-predicted flat after transfer to low volatility. I read this as evidence that the learned class boundaries changed with the data distribution, rather than as a stable market mechanism. The main difficulty is that returns are noisy, labels meet at a learned threshold, and both asset behaviour and class balance shift over time.

Limitations and future work. Three seeds show optimization variability but not sampling uncertainty. I tested only two assets and one future month, with just 75 high-volatility July samples. Technical indicators omit order-book, macroeconomic, and sentiment information. Future work should register multiple future months, add confidence intervals or block-bootstrap tests, study calibration, and compare rolling retraining under a pre-declared protocol. Any trading study would separately require fees, slippage, latency, and risk controls.

9 Ethical Considerations

Forecasts can encourage overconfidence, speculative losses, or unequal harm to users who cannot absorb volatility. Binance data may not represent other venues, outages, liquidity, or manipulation; historical patterns can fail abruptly. The system must be presented as an educational robustness study, with uncertainty and failure cases visible, not as financial advice. Reproducible frozen artifacts reduce selective reporting, but they do not make the predictions safe for deployment.

References

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